

Some Pre-Complex Analysis: Motion on the Unit Circle and Roots of Unity

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§1.1 The complex exponential as motion

The first example to keep in mind is

$$Z(t) = e^{it}.$$

Here t is time and $Z(t)$ is the position of a moving particle in the complex plane.

Differentiating gives

$$\frac{d}{dt}e^{it} = ie^{it}.$$

Thus the velocity is

$$V(t) = iZ(t).$$

Multiplication by i rotates a complex number by a right angle, so the velocity is always perpendicular to the position vector.

At $t = 0$, the particle starts at

$$Z(0) = 1,$$

and the velocity is

$$V(0) = i.$$

So the particle initially moves upward. A moment later the position vector has turned slightly, and the velocity is again obtained by rotating that new position vector by 90° . Continuing this construction gives motion around the unit circle.

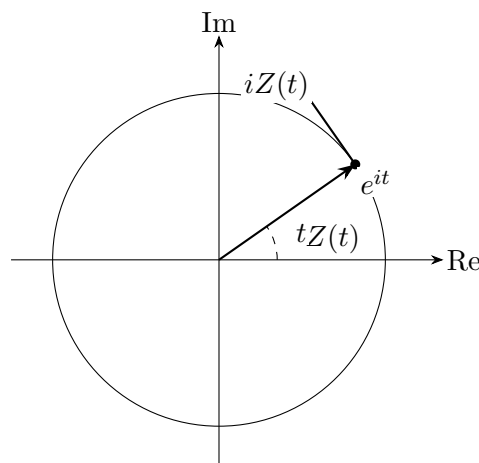


Figure 1.1: For $Z(t) = e^{it}$, multiplication by i rotates the position vector by 90° , giving the tangent velocity.

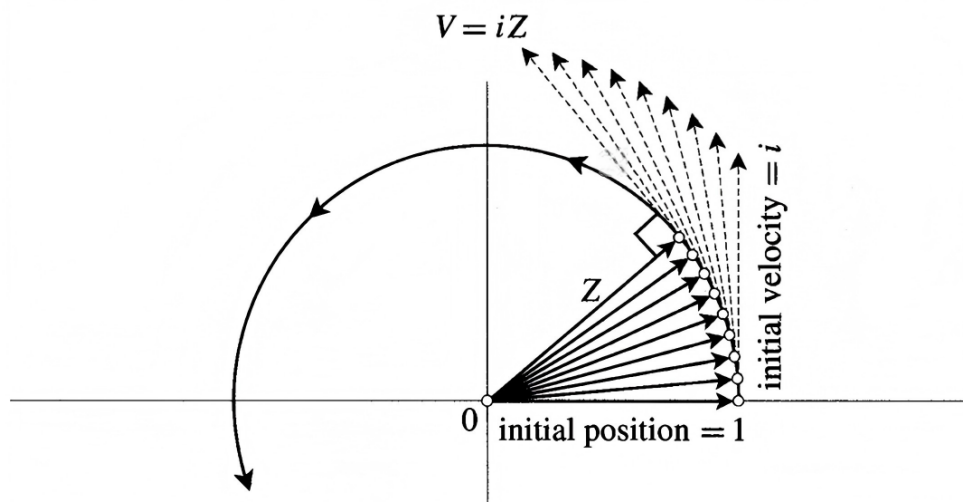


Figure 1.2: Original geometric picture for $Z'(t) = iZ(t)$: the velocity is obtained by multiplying the current position by i , so the motion follows the unit circle from the initial position 1.

§1.2 Roots and angles

The same geometric viewpoint is useful for roots in the complex plane. A root of unity is best understood not merely as a solution of an equation, but as a point on the unit circle with a specified angle.

Remark 1.2.1 — This is why complex numbers make polynomial divisibility questions visual. If a polynomial is built from powers of x , its roots often arrange themselves by angles on the unit circle.

§1.3 A divisibility result

The note records a result analogous to a previous observation about roots and angles.

Theorem 1.3.1

The polynomial

$$x^{2n} + x^n + 1$$

has the factor

$$x^2 + x + 1$$

if and only if

$$n \not\equiv 0 \pmod{3}.$$

Proof. The roots of $x^2 + x + 1$ are the primitive third roots of unity, say

$$\omega = e^{2\pi i/3}, \quad \omega^2 = e^{4\pi i/3}.$$

Thus $x^2 + x + 1 \mid x^{2n} + x^n + 1$ exactly when

$$\omega^{2n} + \omega^n + 1 = 0.$$

Let $k \equiv n \pmod{3}$. If $k = 0$, then $\omega^n = 1$, so the value is

$$1 + 1 + 1 = 3 \neq 0.$$

If $k = 1$, then $\omega^n = \omega$, and

$$\omega^{2n} + \omega^n + 1 = \omega^2 + \omega + 1 = 0.$$

If $k = 2$, then $\omega^n = \omega^2$, and

$$\omega^{2n} + \omega^n + 1 = \omega^4 + \omega^2 + 1 = \omega + \omega^2 + 1 = 0.$$

Therefore divisibility holds precisely when $n \not\equiv 0 \pmod{3}$. □

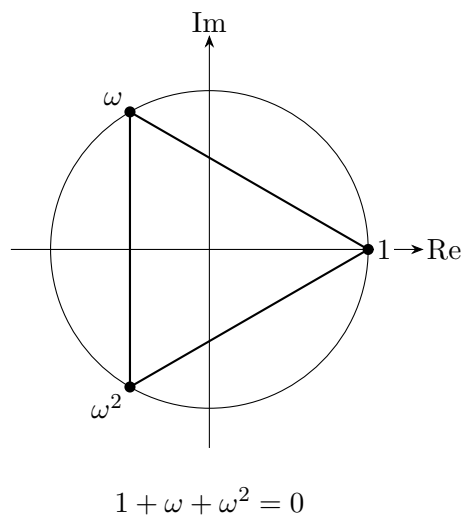


Figure 1.3: Primitive third roots of unity on the unit circle. Divisibility by $x^2 + x + 1$ is checked by evaluating at ω and ω^2 .

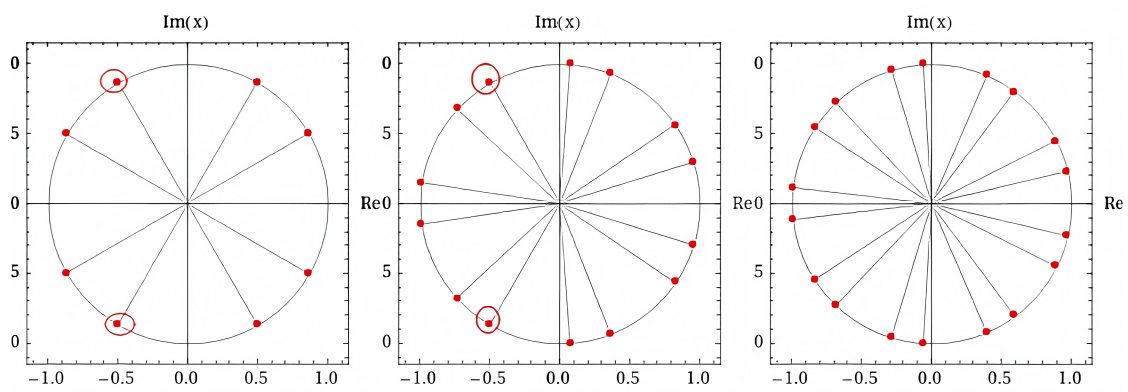


Figure 1.4: Original root-of-unity plots: increasing the number of roots fills the unit circle more densely, while highlighted roots show how special powers or congruence classes are read geometrically.

§1.4 Closing perspective

The two ideas in this note are connected by geometry. The derivative of e^{it} says that motion on the unit circle is governed by rotation by i . The divisibility theorem says that evaluating a polynomial at roots of unity turns algebra into angle arithmetic.

§1.5 Why multiplication by i is rotation

Write a complex number as $z = x + iy$. Then

$$iz = i(x + iy) = -y + ix.$$

As a vector, this sends

$$(x, y) \mapsto (-y, x),$$

which is rotation by 90° counterclockwise. Therefore the differential equation

$$Z'(t) = iZ(t)$$

says that the velocity is always obtained by rotating the radius vector by a right angle. This is exactly uniform circular motion.

The equation also explains why exponentials appear in complex analysis: the function e^{it} is the solution of a linear differential equation whose coefficient is multiplication by i .

§1.6 Roots of unity as symmetry

The equation

$$z^n = 1$$

has solutions

$$z_k = e^{2\pi i k/n}, \quad k = 0, 1, \dots, n-1.$$

These points form a regular n -gon on the unit circle. Multiplying by $e^{2\pi i/n}$ rotates the polygon by one step. Thus roots of unity are not merely algebraic solutions; they are the rotational symmetries of the circle.

For $n = 3$, if $\omega = e^{2\pi i/3}$, then

$$1 + \omega + \omega^2 = 0.$$

Geometrically, the three vectors form an equilateral triangle centered at the origin, so their sum is zero. Algebraically, the same identity comes from

$$x^3 - 1 = (x - 1)(x^2 + x + 1).$$

§1.7 Context and outlook

This note is a prelude to complex analysis because it already contains three central ideas: complex multiplication is geometry, exponentials solve differential equations, and roots of unity connect algebraic equations with symmetry. Later topics such as residues, Fourier analysis, cyclotomic polynomials, and covering maps all reuse this same unit-circle picture.

§1.8 Roots of unity as a finite symmetry group

The n -th roots of unity form a cyclic group

$$\mu_n = \{z \in \mathbb{C} : z^n = 1\} = \{e^{2\pi i k/n} : 0 \leq k < n\}.$$

Multiplication corresponds to addition of exponents modulo n . Thus roots of unity are not just solutions of an equation; they are the rotational symmetries of a regular n -gon.

§1.9 Cyclotomic factorization

The factorization

$$x^n - 1 = \prod_{d|n} \Phi_d(x)$$

separates roots of unity by exact order. The polynomial Φ_n collects primitive n -th roots. This is the algebraic form of decomposing rotations by their period.

Such factorization explains why divisibility problems involving $x^n - 1$ often reduce to order and gcd computations.

§1.10 Fourier viewpoint

Roots of unity also underlie the discrete Fourier transform. For a sequence $(a_0, \dots, a_{n-1})$, the transform evaluates the polynomial

$$A(x) = \sum_{k=0}^{n-1} a_k x^k$$

at roots of unity. Orthogonality of characters gives inversion.

§1.11 Cyclic symmetry on the unit circle

Motion on the unit circle, roots of unity, and divisibility of cyclotomic expressions are one story: the circle group has finite cyclic subgroups, and algebra records their periods through cyclotomic polynomials and Fourier characters.

§1.12 Roots of unity as characters

The n -th roots of unity form a cyclic group

$$\mu_n = \{z \in \mathbb{C} : z^n = 1\}.$$

If $\zeta = e^{2\pi i/n}$, then every root is ζ^k . The functions

$$\chi_m(k) = \zeta^{mk}$$

are characters of the finite cyclic group $\mathbb{Z}/n\mathbb{Z}$.

Orthogonality of roots of unity becomes character orthogonality:

$$\sum_{k=0}^{n-1} \zeta^{mk} = 0$$

unless $n \mid m$, in which case the sum is n .

§1.13 Discrete Fourier transform

For a function $f : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{C}$, its discrete Fourier transform is

$$\widehat{f}(m) = \sum_{k=0}^{n-1} f(k) e^{-2\pi i m k / n}.$$

The inverse formula is

$$f(k) = \frac{1}{n} \sum_{m=0}^{n-1} \widehat{f}(m) e^{2\pi i m k / n}.$$

Thus roots of unity are the basis functions for finite periodic data.

This connects elementary complex-number geometry to signal processing, finite harmonic analysis, and representation theory of finite abelian groups.

§1.14 Line-to-circle covering map

The exponential map

$$\mathbb{R} \rightarrow S^1, \quad t \mapsto e^{it}$$

is a covering map with kernel $2\pi\mathbb{Z}$. It wraps the real line around the circle infinitely many times.

Angles modulo 2π are therefore elements of the quotient group

$$\mathbb{R}/2\pi\mathbb{Z}.$$

The elementary fact that angles differing by 2π are equivalent is a quotient-space statement.

§1.15 Cyclotomic fields

The field $\mathbb{Q}(\zeta_n)$ generated by a primitive n -th root of unity is a cyclotomic field. Its Galois group is

$$\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$

The elementary root powers ζ_n^a become field automorphisms when a is invertible modulo n .

Thus roots of unity are simultaneously geometry of the circle, Fourier basis functions, and algebraic numbers with arithmetic symmetries.

§1.16 Finite Fourier analysis from roots of unity

The orthogonality identity

$$\sum_{k=0}^{n-1} \zeta^{mk} = 0$$

for $n \nmid m$ is the engine of finite Fourier analysis. It implies that the matrix

$$F_{mk} = \zeta^{mk}$$

is unitary up to normalization. Therefore every function on a cyclic group decomposes uniquely into frequency components.

This gives an advanced interpretation of elementary root-of-unity sums. They are not only tools for solving polynomial equations; they are projection formulas onto irreducible characters of a finite abelian group.

§1.17 Cyclotomic fields and constructibility

Roots of unity also control constructible regular polygons. A regular n -gon is constructible by straightedge and compass exactly when the relevant cyclotomic field can be built by successive quadratic extensions. Gauss' theorem says this occurs when

$$n = 2^k p_1 \cdots p_m$$

where the p_i are distinct Fermat primes.

Thus elementary geometry of regular polygons connects to field extensions and Galois theory through roots of unity.